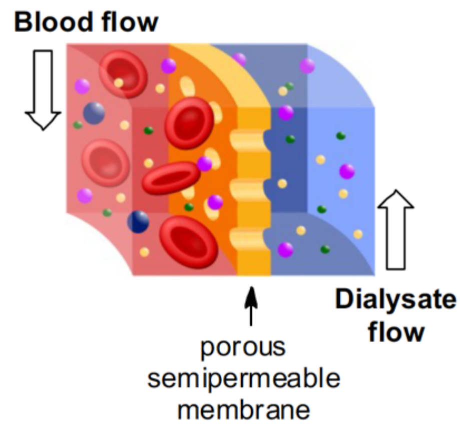
CE318 Team Project 3: Hemodialysis

(Each team should submit one hard copy of project report. Due date: Thursday April 30)

Hemodialysis is a membrane separation process to remove urea and other waste materials from the bloodstream of patients whose kidneys are not functioning properly. During hemodialysis, blood to be treated is run inside hollow fibers composed of a membrane with pore sizes that allow urea and other wastes to diffuse out of the blood while restricting macromolecules (such as proteins, etc.) and blood cells. Dialysate, a water solution containing make-up electrolytes to replace those lost from the blood stream, is flushed outside the fibers counter-currently to the blood flow.



The information regarding operating conditions is provided below:

Flow rate and concentration data:

Urea concentration in bloodstream of patient requiring dialysis (X_{bi}) = 2000 mg/L
Target concentration of urea in bloodstream after treatment (X_{bo}) = 600 mg/L
Flow rate of blood in mL/min (V_b) = 1000 mL/min
Flow rate of blood in m/s (υ_b): $\upsilon_d V_b/2V_d$
Flow rate of dialysate in mL/min (V_d) = 2400 mL/min
Flow rate of dialysate in m/s (υ_d) = 0.2 m/s

Equipment specifications:

Operation temperature: 37.4 ºC. ; Fiber length: 20 cm
Membrane thickness = 25 microns. ; Open void fraction of membrane = 0.2
Diffusivity of urea in water at 26.7 ºC = 1.44 Å~ 10^{-9} m^2/s

In addition to other assumptions you may choose to take, consider the following:

1) There are 45 vol% of blood cells and 55 vol% of plasma in blood (ρ = 990 kg/m^3, μ = 3.0 cp). Plasma has water as the major component (>90%), with a protein concentration of ~75 g/L. About 30% of urea may be bound with protein. An average molecular weight of 100,000 kg mass/kg mol can be assumed for protein.
2) Because the thickness of the membrane is much smaller than fiber diameter, you may assume that the membrane is essentially "flat".
3) You may assume that dialysate has the same properties as water under same conditions.
4) In practice, this is an unsteady state problem since the concentration of urea entering the unit would drop as the patient receives treatment. However, as a first order approximation you may assume steady-state behavior.

Please determine the surface area of fibers needed for a hemodialysis unit and estimate the cost.

Please show all equations and all calculation steps to get the final answer (50% pts for equations and justifications; 50% pts for calculations and results).